

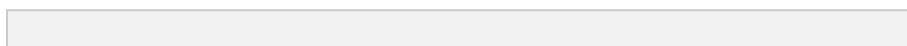
Food of thoughts.

..and another thought

Obviously, an algorithm should not only use multiplications and weights, but also consider conditionality.

- For example, if the same compound is found with multiple adducts, that strengthens the overall probability.
- However, if a compound is found only with a single in-source fragment, and then even a weird or uncommon one, that should weaken the probability.

Here's an example for a compound I had flagged because of a deltaRTpred -44s



but now I will move it to unknowns because of that uncommon ISF

Oliver

On Tue, Mar 31, 2026 at 9:51 AM Oliver Fiehn <ofiehn@ucdavis.edu> wrote:

Hi,

a little bit almost as a note to myself (so that I don't forget about it), now cc'ing Fanzhou, we may also glean other derivative information **C and D** into a unified probability score, but clearly steps A and B are more critical.

, here are the steps we need to take:

(a) [Add the number of patents and the number of pubmed papers for all annotated compounds in MassWiki](#), starting e.g. with HILIC negESI. We get those numbers from cts-lite.metabolomics.us

- If a compound has no citations or annotations, it is unlikely to be true.

- The higher the number of PubMed citations is, the higher the likelihood our annotation is true.

(b) **more importantly**, adding the delta MSMS difference to the next-best-hit from MassWiki library searches for all annotated compounds. (If there is no next-best-hit, that's a strong indication that

(c) adding a probability measure for **low library MSMS entropy**. So far we only consider experimental MSMS entropy, but if the library MSMS has low information content, that should also lower the overall probability

(d) **reverse similarity**. Usually, we match experimental MSMS to library MSMS. That should also stay the case. But I noticed that when I look at hits manually, I also consider reverse score similarity in my thoughts, i.e. are all library MSMS fragments also found in the experimental MSMS? That could be another contribution to the probability score, because we know that we still have quite a bit of noise in experimental MSMS spectra....

best

Oliver

On Tue, Mar 31, 2026 at 8:48 AM Oliver Fiehn <ofiehn@ucdavis.edu> wrote:

Hi Ziyue,

- Basically, we need to move from statistics to data science.

The idea is to gain insights ourselves, before telling Quentin what algorithms to engineer into MassWiki. Alternatively, we can ask Claude to find a slick way to combine information, match True Positives from MassWiki or identify likely False Positives (i.e., things I did wrong).

- Attached is an Excel sheet that I am currently working on to manually weed out clear false positives manually from my former annotations.

Instead of classifying MSMS spectra into chemical classes, we should focus on understanding how to integrate the data. I am currently at row #1298, which gives you a peek into my current approach.

Our task (i.e. mine, your's and Joe's, plus Fanzhou and Quentin) is to develop an algorithm that best describes a weighted formula to generate a probability score. It includes

1. precursor MS,
2. [MSMS identity search](#)
3. [MSMS entropy](#)
4. deltaMSMS to the next-best-hit in MassWiki identity searches, (less likely if there are hits to other SMILES within plusminus 100 similarity units)
5. [deltaRT](#)
6. biological probability (gleaned from the number of PubMed and patent hits),
7. the existence of other hits in MassWiki (more likely if a hit has been found at the same RT plusminus 5s with another adduct, less likely if there is another hit in a specific MassWiki DB with the same adduct but a different RT)
8. peak shape (we don't have that information yet automatically...will get it at some point from Gert)

In column AF I used a [very simplistic equation](#) just to draw my own attention to 'possibly false positive annotations'. This equation multiplies the (range-scaled) [delta RT * identity search * entropy](#) (with some weight applied), and adding 0.1 unit to prevent entropy 0 from automatically resulting in a zero probability during multiplication. In column AK I provide my reasoning for moving a compound to unknowns (yy_) or for keeping it despite apparently low probability.

Unfortunately, the MassWiki downloads do not include all the information we need, but maybe you have better access or can work with Quentin to get better access and glean that information?

I wanted a real team meeting, but since the team is not fully here yet (Quentin is working from home today, Joe is still in Spain, returning to the office on Thursday), **here are the steps we need to take:**

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(b) **more importantly**, adding the delta MSMS difference to the next-best-hit from MassWiki library searches for all annotated compounds. (If there is no next-best-hit, that's a strong indication that

thanks

Oliver

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